

A CONSTANT-COEFFICIENT DIRICHLET CASE OF THE ANGLE-COINCIDENCE QUESTION

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SOURCE AND STATUS

This packet concerns arXiv:2510.23450, Meinlschmidt–Rehberg, *Sharp angle estimates for second order divergence operators* [1]. In the final comments, the source asks for useful information on when the inequality

$$\omega(\mathfrak{t}_V) \leq \omega(\mu)$$

is sharp. The source gives a two-dimensional constant skew-symmetric Dirichlet example where the inequality is strict, and then says that a precise useful characterization seems hard.

$$\omega(\mathfrak{t}_V) \leq \omega(\mu) \leq \operatorname{ess\,sup}_{x \in \Omega} \frac{n(\Im \mu(x))}{m(x)}.$$

We have already discussed sharpness of the second inequality in Remark 2.4, yet, of course, it is clearly also very interesting to understand whether or when the first one is sharp as well. In fact, there is a generic way to generate examples for which $\omega(\mathfrak{t}_V) < \omega(\mu)$: choose V and a coefficient matrix μ such that μ is not selfadjoint, but the operator A induced by the form $\mathfrak{t}_V$ is so. Such a construction seems to be hinted at in [21, Rem. 4]. Then $N(A)$ is real, thus so is $N(\mathfrak{t}_V)$ and we obtain $\omega(\mathfrak{t}_V) = 0$. But $N(\mu)$ cannot be a subset of $\mathbb{R}$, otherwise μ would be selfadjoint, so that $\omega(\mu) > 0$. Let for example $V = W_{\partial\Omega}^{1,2}(\Omega) = W_0^{1,2}(\Omega)$ with an open set $\Omega \subseteq \mathbb{R}^2$ and pick, say, $\mu = \begin{pmatrix} 1 & -\alpha \\ \alpha & 1 \end{pmatrix}$ with $\alpha \in \mathbb{R}$. Then A becomes the Dirichlet Laplacian. As mentioned before, such examples depend on both the coefficient matrix and the subspace V . Indeed, for $V = W^{1,2}(\Omega)$ with Ω bounded (in any space dimension), the example does not work and one can show that $\omega(\mathfrak{t}) = \omega(\mu)$, see [2, Rem. 5.12]. It seems however quite hard to give a precise and useful characterization on when the angles coincide or not. In one space dimension it is not hard to see that there is always equality, but this relies crucially on the particular dimension 1. We leave further investigations and partial results in this direction for future work.

The result below supplies a small but exact characterization in the constant real full-Dirichlet class.

DEFINITIONS

Let $\Omega \subseteq \mathbb{R}^d$ be a nonempty open set and set $V = W_0^{1,2}(\Omega)$. For a constant real matrix $M \in \mathbb{R}^{d \times d}$ whose symmetric part is uniformly positive, consider the Dirichlet form

$$\mathfrak{t}(u, v) = \int_{\Omega} (M \nabla u, \nabla v)_{\mathbb{C}^d} dx, \quad u, v \in W_0^{1,2}(\Omega).$$

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Write

$$M = S + K, \quad S = \frac{M + M^T}{2}, \quad K = \frac{M - M^T}{2}.$$

Thus $S = S^T$ and $K^T = -K$. Assume $S \geq mI$ for some $m > 0$.

PROOF INTUITION

With full Dirichlet boundary conditions, a constant skew-symmetric coefficient matrix contributes only a divergence-free second-order expression. Integration by parts kills it in the weak form. The form angle is therefore controlled only by the symmetric positive part S , so it is zero. The coefficient angle, however, is computed from the numerical range of the matrix itself; a nonzero real skew-symmetric part produces a nonzero imaginary numerical direction. This separates the two angles exactly.

THEOREM

Let $\Omega \subseteq \mathbb{R}^d$ be a nonempty open set and let $M = S + K \in \mathbb{R}^{d \times d}$ be constant with $S = S^T \geq mI$ and $K^T = -K$. For the full Dirichlet form on $V = W_0^{1,2}(\Omega)$,

$$\omega(\mathfrak{t}_V) = 0.$$

Moreover, the coefficient angle satisfies $\omega(M) = 0$ if and only if $K = 0$. Consequently, in this class,

$$\omega(\mathfrak{t}_V) = \omega(M) \iff M \text{ is symmetric.}$$

Proof. First suppose $u, v \in C_c^\infty(\Omega)$. Since K is constant and skew-symmetric,

$$\int_{\Omega} (K \nabla u, \nabla v)_{\mathbb{C}^d} dx = \sum_{j,k=1}^d K_{jk} \int_{\Omega} \partial_k u \overline{\partial_j v} dx.$$

Integrating by parts in the j variable gives

$$- \sum_{j,k=1}^d K_{jk} \int_{\Omega} \partial_j \partial_k u \bar{v} dx = 0,$$

because the matrix $(\partial_j \partial_k u)_{j,k}$ is symmetric while K is skew-symmetric. By density of $C_c^\infty(\Omega)$ in $W_0^{1,2}(\Omega)$, the identity

$$\int_{\Omega} (K \nabla u, \nabla v)_{\mathbb{C}^d} dx = 0$$

holds for all $u, v \in W_0^{1,2}(\Omega)$. Hence

$$\mathfrak{t}(u, u) = \int_{\Omega} (S \nabla u, \nabla u)_{\mathbb{C}^d} dx \in [0, \infty)$$

for every $u \in W_0^{1,2}(\Omega)$, and therefore the numerical range of the form is contained in the positive real axis. Thus $\omega(\mathfrak{t}_V) = 0$.

It remains to identify when the coefficient angle is zero. If $K = 0$, then $M = S$ is selfadjoint and positive, so $N(M) \subset [0, \infty)$ and $\omega(M) = 0$. Conversely, assume $K \neq 0$. The operator

$$\text{ImOp}(M) = \frac{M - M^T}{2i} = -iK$$

is a nonzero selfadjoint operator on $\mathbb{C}^d$, hence has a nonzero real eigenvalue λ with unit eigenvector ξ . Then

$$\text{Im}(M\xi, \xi)_{\mathbb{C}^d} = (\text{ImOp}(M)\xi, \xi)_{\mathbb{C}^d} = \lambda \neq 0,$$

whereas

$$\text{Re}(M\xi, \xi)_{\mathbb{C}^d} = (S\xi, \xi)_{\mathbb{C}^d} > 0.$$

Thus $N(M)$ is not contained in the positive real axis, and $\omega(M) > 0$. This proves the equivalence. $\square$

VERIFICATION NOTES AND SCOPE

The proof is purely analytic. It uses only density of $C_c^\infty(\Omega)$ in $W_0^{1,2}(\Omega)$ and the elementary spectral fact that a nonzero selfadjoint finite-dimensional operator has a nonzero numerical direction.

The result is intentionally narrow. It gives a complete characterization in the constant real full-Dirichlet class, extending the source paper's displayed two-dimensional skew example, but it does not handle variable coefficients, mixed or Neumann boundary conditions, or the complex-coefficient p -numerical-range problem raised in the next item of the source comments.

HUMAN REVIEW RECOMMENDATION

Review as a scoped partial result toward the angle-coincidence question. The main point to check is the integration-by-parts cancellation of the constant skew-symmetric part in the full Dirichlet form.

REFERENCES

- [1] H. Meinschmidt and J. Rehberg, *Sharp angle estimates for second order divergence operators*, arXiv:2510.23450.